

**RAFI KAMIL ARIEF****Electrical Engineering Graduate | Technology & Systems Engineering**

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<https://www.linkedin.com/in/rafikamilarief>**SUMMARY**

Electrical Engineering graduate from the University of Indonesia currently pursuing an accelerated Master's degree in Electronic and Intelligent Embedded Systems. Focused on FPGA-based embedded control, digital system implementation, and industrial electronics, with research and practical exposure in sensor systems, hardware experimentation, and engineering analysis. Experienced in academic assistance, technical coordination, and cross-functional collaboration through research, internship, and organizational activities related to energy and engineering systems.

EDUCATION**University of Indonesia**, West Java, Indonesia August 2025 – Present*MEng in Electronic and Intelligent Embedded System*

- Accelerated Bachelor-Master Program.
- GPA: 3.95 / 4.00
- Focus: FPGA-based embedded systems and digital control implementations.

University of Indonesia, West Java, Indonesia

July 2021 – July 2025

BEng in Electrical Engineering

- GPA: 3.76 / 4.00
- Thesis: Fiber Bragg Grating (FBG)-based hydrostatic pressure sensing system using finite element modeling and experimental validation

WORK & RESEARCH EXPERIENCE**University of Indonesia**, West Java, Indonesia

March 2023 – Present

Assistant Lecturer

- Assisted teaching activities for electric circuits, semiconductor devices, and electronic circuits courses.
- Supported problem-solving sessions and engineering concept discussions for undergraduate students.

Bimbingan Tes Alumni (BTA) 45, Jakarta, Indonesia

August 2025 – December 2025

Physics Tutor

- Mentored high school physics with emphasis on conceptual understanding, problem-solving strategies, and exam-oriented preparation.
- Guided students in translating physical principles into structured mathematical solutions.

National Research and Innovation Agency (BRIN), Banten, Indonesia

August 2024 – July 2025

Undergraduate Thesis Researcher

- Developed diaphragm-based Fiber Bragg Grating (FBG) sensor system for hydrostatic pressure measurement.
- Conducted experimental characterization and engineering data analysis using Python and ANSYS.
- Performed simulation-assisted validation between theoretical and experimental models.

PLN Indonesia Power Services, West Java, Indonesia

July 2024 – December 2024

Internship on Project Construction & Maintenance Services Unit

- Coordinated a five-member technical team for Analog Signature Analysis (ASA)-based PCB diagnostics.
- Utilized FT100 and PFL780 instruments to diagnose identify fault signatures.
- Prepared structured technical reports for operational evaluation and management review.

Department of Electrical Engineering FTUI, West Java, Indonesia

October 2023 – September 2024

Internship on Virtual Museum Coordination Project

- Coordinated a 10-member multidisciplinary team responsible for content development and project execution of a departmental virtual museum.
- Led interviews with alumni, lecturers, and professors, and documented departmental milestones for publication in a historical book.

PUBLICATION

Development of a Fiber Bragg Grating-Based Hydrostatic Pressure Sensor Using a Stainless-Steel Diaphragm for Underwater Applications.

Arief, R. K., et al. (2026). Springer Lecture Notes in Electrical Engineering (LNEE), Proceedings of 6th International Conference on Nano Electronics Research and Education (ICNERE).

Presented in Hamamatsu, Shizuoka, Japan.

ORGANIZATIONAL EXPERIENCE

Ikatan Mahasiswa Pascasarjana Fakultas Teknik UI (IMPASTEK), West Java, Indonesia

Secretary of Public Relations (Humas)

January 2025 – Present

- Supported stakeholder engagement and event operations for Indonesia Youth Energy 2025 and IMPASTEK FUNRUN 2025, contributing to program coordination and participant management.

Student Legislative Council (MPM FT UI), West Java, Indonesia

Deputy Head of Commission for Financial Audit and Funding

December 2023 – January 2025

- Led 11 commission members in financial audits for 20+ faculty-level student organizations, covering annual financial planning, monthly cash flow monitoring, and financial performance index evaluation.
- Coordinated with student organizations to improve reporting compliance and audit readiness.
- Implemented workload-based liaison officer allocation to balance audit responsibilities across members while maintaining consistent organizational support.
- Initiated financial governance training and senior-led knowledge sharing sessions to improve internal audit capability and mentoring quality.
- Increased annual financial planning implementation success rates from below 80% to above 80%, exceeding previous administrations.

Member of Commission for Financial Audit and Funding

December 2022 – December 2023

- Supported financial governance across faculty-level student organizations by monitoring monthly cash flow reports and providing direct reporting guidance
- Streamlined monthly reporting procedures, increasing cash flow implementation rates from approximately 60% to above 80% throughout the administrative year.
- Served as liaison officer for the Electrical Engineering Student Association, contributing to a 35% improvement in its financial performance index compared to the previous administration.

VOLUNTEER EXPERIENCE

Electrical Engineering Student Association (IME FT UI), West Java, Indonesia

July 2023 – July 2024

Freshmen's Mentor

- Mentored 10 incoming Electrical Engineering students, providing academic guidance, adaptation support, and orientation throughout their first year.

NEST UI, West Java, Indonesia

July 2022 – August 2022

Deputy Head of Publication Division

- Coordinated digital promotion initiatives for a national scientific writing competition, supporting participant outreach and event visibility.

ACHIEVEMENTS

- Top 60 Finalist — Paragon IN:PACT International Business Case Competition 2025**
- Roberto Rocca Scholarship Awardee 2025**

SKILLS

Embedded & Digital Systems

FPGA Design, Verilog, Xilinx Vivado/Vitis, Arduino

Programming

C, Python, HDL, MATLAB

Engineering & Simulation

Cadence Virtuoso, LTSpice, ANSYS, Multisim,

Proteus, EMU8086

CERTIFICATE

FPGA Architecture Based System for Industrial Applications (L&T EduTech, 2026)

TOEFL ITP: 610/677 (ETS, 2026)